Slide 1: Title, introduction, our names.

Slide 2 (The Mess): Before we could analyze anything, we had to find what was wrong with the raw data. Strawberry Pop-Tarts showed up under four different SKU codes, so one product looked like four. Our 14 store records were really only 12 stores, because FL-239 and FL-287 each appeared twice under mismatched IDs. And 884 of our 25,532 transactions had no timestamp at all. Left alone, these would have split demand, double-counted stores, and blurred our surge window. Henna will walk through how we fixed them.

Slide 3 (The Pipeline): We started with over 25,000 transaction records, but the data wasn't ready for analysis in the beginning. We removed 30 exact duplicates, standardized store IDs and timestamps, consolidated 37 inconsistent category labels into 10 categories, and mapped 45 SKU variations into 20 actual products.
One important decision was how to handle the 884 transactions with missing timestamps. Instead of deleting them or assigning dates we couldn't verify, we kept them in our cleaned dataset and removed them from time-based analyses. We then validated row counts before using the cleaned data for our analyses

Slide 4 (Q1): For our first question, we wanted to identify which product categories experienced the biggest increase in demand leading up to Hurricane Helene. We compared average daily sales from our baseline period, September 1st through 15th, to the seven days before landfall, September 19th through 25th. What we found was that Breakfast Foods had the largest increase in demand, rising 5.14 times, or about 414%. Average daily sales jumped from around 73 units to 377 units. This was especially interesting because while we expected emergency essentials to increase, breakfast foods actually experienced the biggest surge. It shows how customers prepare for hurricanes by purchasing convenient, shelf-stable foods, not just traditional emergency supplies.

Slide 5 (Q2): For SKU consolidation, we discovered that 45 different identifiers actually represented only 20 products. For example, Strawberry Pop-Tarts appeared under 4 different SKUs. Individually, the largest SKU showed only 1,377 units, but combining them reveal 5,178 units for the same product. This is important because fragmented identifiers could cause Walmart to underestimate demand and make poor inventory decisions. We also verified our consolidation didn't change total counts.

Slide 6 (Q3): For Question 3, we analyzed how weather conditions during Hurricane Helene were associated with Walmart sales across 12 stores. Our first finding was a sharp increase in sales before the hurricane, rising from approximately 1,500 to nearly 4,000 units, suggesting possible emergency stockpiling. Sales then dropped significantly around landfall on September 26. We also found that previous-day rainfall had the strongest negative correlation with sales, at -0.55, suggesting a potential delayed relationship. Wind speed and atmospheric pressure showed weaker correlations. Overall, our findings suggest increased purchasing before the storm and reduced shopping activity during landfall, although correlation does not establish causation.

Slide 7 (Q4): Not one SKU: Monopoly rose 2.60x and playing cards 1.99x, each on its own.
Not one store: all 12 stores rose, and the top store was only 10.1% of surge-week units.
Matches a real signal: Entertainment tracked Breakfast day by day (r = 0.98 across 30 days).
Recommendation: pilot a modest board-game increase at selected stores before the next storm.

The SQLite database shows the same surge (about 2.5x, peaking Sept 24–25), but its volumes don’t match the transactions (0 of 338 store-days), so we’re not yet confident enough to stock at scale.

Slide 8 (Q5): For question 5, the case asks whether the board game spike is real or an artifact. Entertainment rose 2.25 times in the week before landfall, so we tested it three ways. Monopoly rose 2.6 times and playing cards 1.99 times, so it is not one bad SKU. All 12 stores rose, and the top store was only 10 percent of the surge, so it is not one store. And Entertainment tracked Breakfast, our confirmed signal, with a correlation of 0.98 across 30 days. Our verdict is signal. One caveat: our SQLite cross-check did not match the transactions, so we recommend a modest pilot, not a full rollout.

Slide 9 (The Recommendation): Based on our findings, I would recommend Walmart prioritize Pop-Tarts, since demand increased about 5.1 times, followed by emergency supplies, which had larger increases in actual units sold. Rather than stocking exactly 5.14 times normal demand, we recommend planning for around 4–5 times and beginning shipments to all 12 stores about a week before landfall. Walmart should also monitor weather forecasts to anticipate purchasing spikes. For board games, we recommend a smaller inventory increase initially, then monitoring sales before making larger stocking decisions.

Slide 10 (What Broke & What Mattered): One challenge we encountered was missing geographic coordinates for our Tallahassee Walmart store. Since our analysis depended on matching each store to historical weather data, missing coordinates meant we couldn't retrieve weather observations for that location. Instead of excluding the store, we used Tallahassee's approximate city-center coordinates as a proxy and documented that assumption. This taught us the importance of validating data before analysis, because even one missing location can affect the completeness of the dataset and potentially influence our results.